APK SBOM Recovery Report

`data/test_apks/real/F-Droid.apk`

Summary

Maven components **23**	Native files **4**	DEX files analyzed **2**
DEX parse failures **0**	Extraction risk **low**	Obfuscation risk **low**
Vulnerability lookup **0**		

APK

SHA-256	`985f5181d48bb6bafd54083a048b391271e0ab28385881cc41294fb01a222762`
PACKAGE	`org.fdroid.fdroid`
VERSION	1.23.2 / code 1023052
SDK	min 23 / target 30
MANIFEST PARSE	parsed
SIZE	12426276 bytes
MULTIDEX	true
MANIFEST	true
RESOURCES	true

APK Forensics

STANDARD DEX	classes.dex, classes2.dex
NONSTANDARD DEX	none
NATIVE ABIS	arm64-v8a, armeabi-v7a, x86, x86_64
PERMISSIONS	android.permission.ACCESS_COARSE_LOCATION, android.permission.ACCESS_NETWORK_STATE, android.permission.ACCESS_WIFI_STATE, android.permission.BLUETOOTH, android.permission.BLUETOOTH_ADMIN, android.permission.BLUETOOTH_CONNECT, android.permission.BLUETOOTH_SCAN, android.permission.CAMERA, android.permission.CHANGE_NETWORK_STATE, android.permission.CHANGE_WIFI_MULTICAST_STATE, android.permission.CHANGE_WIFI_STATE, android.permission.ENFORCE_UPDATE_OWNERSHIP, android.permission.FOREGROUND_SERVICE, android.permission.INTERNET, android.permission.MANAGE_EXTERNAL_STORAGE, android.permission.NFC, android.permission.POST_NOTIFICATIONS, android.permission.QUERY_ALL_PACKAGES, android.permission.READ_EXTERNAL_STORAGE, android.permission.RECEIVE_BOOT_COMPLETED, android.permission.REQUEST_DELETE_PACKAGES, android.permission.REQUEST_INSTALL_PACKAGES, android.permission.UPDATE_PACKAGES_WITHOUT_USER_ACTION, android.permission.USB_PERMISSION, android.permission.WAKE_LOCK, android.permission.WRITE_EXTERNAL_STORAGE, android.permission.WRITE_SETTINGS, org.fdroid.fdroid.DYNAMIC_RECEIVER_NOT_EX

	PORTED_PERMISSION, org.fdroid.fdroid.permission.UPDATE_REPOS
OBFUSCATION RISK	low score=0.20 • At least half of emitted components are low-confidence version candidates.

SEVERITY	CODE	DETAIL
No extraction indicators were raised.		

Detected Components

COMPONENT	VERSION	REASONS	IDENTITY	VERSION CONFIDENCE	MATCH TYPE	VULNS	EVIDENCE
`androidx.gridlayout:gridlayout` pkg:maven/androidx.gridlayout/gridlayout@1.0.0	1.0.0	package, class, method, opcode, graph, literals, version-unique	confirmed/high (0.983)	exact/high (0.972)	hybrid-exact	0	▶ Evidence
`androidx.localbroadcastmanager:localbroadcastmanager` pkg:maven/androidx.localbroadcastmanager/localbroadcastmanager@1.0.0	1.0.0	package, class, method, opcode, graph, literals	confirmed/high (0.962)	candidate/low (0.871)	hybrid-candidate	0	▶ Evidence

COMPONENT	VERSION	REASONS	IDENTITY	VERSION CONFIDENCE	MATCH TYPE	VULNS	EVIDENCE
`androidx.swiperefreshlayout:swiperefreshlayout` pkg:maven/androidx.swiperefreshlayout/swiperefreshlayout@1.0.0	1.0.0	package, class, method, opcode, graph, literals, version-unique	confirmed/high (0.919)	likely/medium (0.870)	hybrid-likely	0	▶ Evidence
`androidx.concurrent:concurrent-futures` pkg:maven/androidx.concurrent/concurrent-futures@1.1.0	1.1.0	package, class, method, opcode, graph, literals	confirmed/high (0.937)	candidate/low (0.843)	hybrid-candidate	0	▶ Evidence
`androidx.recyclerview:recyclerview` pkg:maven/androidx.recyclerview/recyclerview@1.4.0	1.4.0	package, class, method, opcode, graph, literals, version-unique	likely/medium (0.835)	likely/medium (0.832)	hybrid-likely	0	▶ Evidence
`androidx.appcompat:appcompat` pkg:maven/androidx.appcomp	1.7.0	package, class, method, opcode, graph, literals,	likely/medium (0.802)	likely/medium (0.815)	hybrid-likely	0	▶ Evidence

COMPONENT	VERSION	REASONS	IDENTITY	VERSION CONFIDENCE	MATCH TYPE	VULNS	EVIDENCE
at/appcompat@1.7.0		version-unique					
`androidx.lifecycle:lifecycle-viewmodel-android` pkg:maven/androidx.lifecycle/lifecycle-viewmodel-android@2.8.7	2.8.7	package, class, method, opcode, graph, literals, version-unique, artifact-family	likely/medium (0.822)	likely/medium (0.808)	hybrid-likely	0	▶ Evidence
`com.squareup.okhttp3:okhttp` pkg:maven/com.squareup.okhttp3/okhttp@4.12.0	4.12.0	package, class, method, opcode, graph, literals, version-unique	likely/medium (0.802)	likely/medium (0.807)	hybrid-likely	0	▶ Evidence
`androidx.exifinterface:exifinterface` pkg:maven/androidx.exifinterface/exifinterface@1.3.6	1.3.6	package, class, method, opcode, graph, literals, version-unique	likely/medium (0.813)	likely/medium (0.792)	hybrid-likely	0	▶ Evidence
`io.ktor:ktor-client-okhttp-jvm` pkg:maven/io.ktor/ktor-client-	2.3.12	package, class, method, opcode, graph, literals	likely/medium (0.877)	candidate/low (0.788)	hybrid-candidate	0	▶ Evidence

COMPONENT	VERSION	REASONS	IDENTITY	VERSION CONFIDENCE	MATCH TYPE	VULNS	EVIDENCE
okhttp-jvm@2.3.12							
`androidx.work:work-runtime` pkg:maven/androidx.work/work-runtime@2.10.0	2.10.0	package, class, method, opcode, graph, literals, version-unique	likely/medium (0.757)	likely/medium (0.752)	hybrid-likely	0	▶ Evidence
`androidx.fragment:fragment` pkg:maven/androidx.fragment/fragment@1.6.1	1.6.1	package, class, method, opcode, graph, literals, version-unique	likely/medium (0.763)	candidate/low (0.749)	hybrid-candidate	0	▶ Evidence
`androidx.compose.ui:ui-android` pkg:maven/androidx.compose.ui/ui-android@1.7.8	1.7.8	package, class, method, opcode, graph, literals, version-unique	likely/medium (0.744)	candidate/low (0.727)	hybrid-candidate	0	▶ Evidence
`org.jetbrains.kotlinx:kotlinx-coroutines-android` pkg:maven/org.jetbrains.kotlinx/kotlinx-coroutines	1.7.3	package, class, method, opcode, graph, literals, version-unique	candidate/low (0.674)	candidate/low (0.706)	hybrid-candidate	0	▶ Evidence

COMPONENT	VERSION	REASONS	IDENTITY	VERSION CONFIDENCE	MATCH TYPE	VULNS	EVIDENCE
- android@1.7.3							
`androidx.lifecycle:lifecycle-runtime-android` pkg:maven/androidx.lifecycle/lifecycle-runtime-android@2.8.7	2.8.7	package, class, method, opcode, graph, literals, version-unique, artifact-family	likely/medium (0.707)	candidate/low (0.676)	hybrid-candidate	0	▶ Evidence
`androidx.activity:activity` pkg:maven/androidx.activity/activity@1.10.1	1.10.1	package, class, method, opcode, graph, literals, version-unique	candidate/low (0.648)	candidate/low (0.633)	hybrid-candidate	0	▶ Evidence
`androidx.lifecycle:lifecycle-livedata` pkg:maven/androidx.lifecycle/lifecycle-livedata@2.8.7	2.8.7	package, class, method, opcode, graph, literals, version-unique	candidate/low (0.585)	candidate/low (0.578)	hybrid-candidate	0	▶ Evidence
`com.squareup.okio:okio-jvm` pkg:maven/com.squareup.okio	3.7.0	package, class, method, opcode, graph, literals,	candidate/low (0.584)	candidate/low (0.543)	hybrid-candidate	0	▶ Evidence

COMPONENT	VERSION	REASONS	IDENTITY	VERSION CONFIDENCE	MATCH TYPE	VULNS	EVIDENCE
/okio-jvm@3.7.0		version-unique, artifact-family					
`org.jetbrains.kotlinx:kotlinx-coroutines-core-jvm` pkg:maven/org.jetbrains.kotlinx/kotlinx-coroutines-core-jvm@1.7.3	1.7.3	package, class, method, opcode, graph, literals, version-unique	candidate/low (0.547)	candidate/low (0.535)	hybrid-candidate	0	▶ Evidence
`com.google.android.material:material` pkg:maven/com.google.android.material/material@1.12.0	1.12.0	package, class, method, opcode, graph, literals, version-unique	candidate/low (0.558)	candidate/low (0.535)	hybrid-candidate	0	▶ Evidence
`androidx.compose.foundation:foundation-android` pkg:maven/androidx.compose.foundation/foundation-android@1.7.8	1.7.8	package, class, method, opcode, graph, literals, version-unique	candidate/low (0.534)	candidate/low (0.524)	hybrid-candidate	0	▶ Evidence

COMPONENT	VERSION	REASONS	IDENTITY	VERSION CONFIDENCE	MATCH TYPE	VULNS	EVIDENCE
androidx.core:core pkg:maven/androidx.core/core@1.15.0	1.15.0	package, class, method, opcode, graph, literals, version-unique	candidate/low (0.533)	candidate/low (0.509)	hybrid-candidate	0	▶ Evidence
io.ktor:ktor-http-jvm pkg:maven/io.ktor/ktor-http-jvm@2.3.12	2.3.12	package, class, method, opcode, graph, literals	candidate/low (0.517)	candidate/low (0.456)	hybrid-candidate	0	▶ Evidence

Version Decisions

COMPONENT	SELECTED VERSION	IDENTITY	VERSION	SCORE	GAP	ALTERNATIVES
androidx.gridlayout:gridlayout	1.0.0	confirmed/high (0.983)	exact/high (0.972)	0.972	0.084	1.1.0 (0.889)
androidx.localbroadcastmanager:localbroadcastmanager	1.0.0	confirmed/high (0.962)	candidate/low (0.871)	0.871	0.000	1.1.0 (0.871)
androidx.swiperefreshlayout:swiperefreshlayout	1.0.0	confirmed/high (0.919)	likely/medium (0.870)	0.870	0.189	1.2.0 (0.681)

COMPONENT	SELECTED VERSION	IDENTITY	VERSION	SCORE	GAP	ALTERNATIVES
androidx.concurrent:concurrent-futures	1.1.0	confirmed/high (0.937)	candidate/low (0.843)	0.843	0.002	1.3.0 (0.841)
androidx.recyclerview:recyclerview	1.4.0	likely/medium (0.835)	likely/medium (0.832)	0.832	0.098	1.2.1 (0.733)
androidx.appcompat:appcompat	1.7.0	likely/medium (0.802)	likely/medium (0.815)	0.815	0.092	1.7.1 (0.723)
androidx.lifecycle:lifecycle-viewmodel-android	2.8.7	likely/medium (0.822)	likely/medium (0.808)	0.808	0.222	2.10.0 (0.587)
com.squareup.okhttp3:okhttp	4.12.0	likely/medium (0.802)	likely/medium (0.807)	0.807	0.074	4.10.0 (0.733), 4.9.1 (0.703), 4.9.0 (0.703)
androidx.exifinterface:exifinterface	1.3.6	likely/medium (0.813)	likely/medium (0.792)	0.792	0.094	1.4.2 (0.699)
io.ktor:ktor-client-okhttp-jvm	2.3.12	likely/medium (0.877)	candidate/low (0.788)	0.788	0.788	none
androidx.lifecycle:lifecycle-runtime	2.6.2	likely/medium (0.851)	candidate/low (0.766)	0.766	0.766	none
androidx.work:work-runtime	2.10.0	likely/medium (0.757)	likely/medium (0.752)	0.752	0.152	2.11.2 (0.600)

COMPONENT	SELECTED VERSION	IDENTITY	VERSION	SCORE	GAP	ALTERNATIVES
androidx.fragment:fragment	1.6.1	likely/medium (0.763)	candidate/low (0.749)	0.749	0.071	1.5.4 (0.678), 1.8.9 (0.650)
androidx.compose.ui:ui-android	1.7.8	likely/medium (0.744)	candidate/low (0.727)	0.727	0.195	1.11.2 (0.532)
org.jetbrains.kotlinx:kotlinx-coroutines-android	1.7.3	candidate/low (0.674)	candidate/low (0.706)	0.706	0.146	1.9.0 (0.560)
androidx.lifecycle:lifecycle-viewmodel	2.6.2	likely/medium (0.775)	candidate/low (0.685)	0.685	0.685	none
androidx.lifecycle:lifecycle-runtime-android	2.8.7	likely/medium (0.707)	candidate/low (0.676)	0.676	0.090	2.10.0 (0.586)
androidx.activity:activity	1.10.1	candidate/low (0.648)	candidate/low (0.633)	0.633	0.074	1.8.0 (0.560), 1.13.0 (0.485)
androidx.lifecycle:lifecycle-livedata	2.8.7	candidate/low (0.585)	candidate/low (0.578)	0.578	0.096	2.10.0 (0.482), 2.6.2 (0.428)
com.squareup.okio:okio	2.8.0	candidate/low (0.639)	candidate/low (0.557)	0.557	0.016	2.9.0 (0.540)
com.squareup.okio:okio-jvm	3.7.0	candidate/low (0.584)	candidate/low (0.543)	0.543	0.004	3.6.0 (0.540), 3.0.0 (0.474)

COMPONENT	SELECTED VERSION	IDENTITY	VERSION	SCORE	GAP	ALTERNATIVES
org.jetbrains.kotlinx:kotlinx-coroutines-core-jvm	1.7.3	candidate/low (0.547)	candidate/low (0.535)	0.535	0.079	1.9.0 (0.456)
com.google.android.material:material	1.12.0	candidate/low (0.558)	candidate/low (0.535)	0.535	0.072	1.13.0 (0.463)
androidx.compose.foundation:foundation-android	1.7.8	candidate/low (0.534)	candidate/low (0.524)	0.524	0.211	1.11.2 (0.314)
androidx.core:core	1.15.0	candidate/low (0.533)	candidate/low (0.509)	0.509	0.047	1.18.0 (0.462)
io.ktor:ktor-http-jvm	2.3.12	candidate/low (0.517)	candidate/low (0.456)	0.456	0.456	none
io.ktor:ktor-client-core-jvm	2.3.12	candidate/low (0.513)	uncertain/uncertain (0.442)	0.442	0.442	none
io.ktor:ktor-utils-jvm	2.3.12	candidate/low (0.474)	uncertain/uncertain (0.419)	0.419	0.419	none
org.jetbrains.kotlin:kotlin-stdlib	2.2.10	candidate/low (0.465)	uncertain/uncertain (0.418)	0.418	0.418	none
io.ktor:ktor-io-jvm	2.3.12	candidate/low (0.468)	uncertain/uncertain (0.417)	0.417	0.417	none
androidx.compose.runtime:runtime-android	1.11.2	uncertain/uncertain (0.401)	uncertain/uncertain (0.362)	0.362	0.362	none

COMPONENT	SELECTED VERSION	IDENTITY	VERSION	SCORE	GAP	ALTERNATIVES
androidx.core:core-ktx	1.18.0	uncertain/uncertain (0.334)	uncertain/uncertain (0.248)	0.248	0.248	none
androidx.compose.material3:material3-android	1.4.0	uncertain/uncertain (0.242)	uncertain/uncertain (0.209)	0.209	0.209	none
com.google.code.gson:gson	2.8.5	uncertain/uncertain (0.159)	uncertain/uncertain (0.170)	0.170	0.012	2.8.9 (0.158), 2.8.6 (0.155)
com.squareup.retrofit2:retrofit	2.9.0	uncertain/uncertain (0.151)	uncertain/uncertain (0.167)	0.167	0.019	2.8.1 (0.148)
androidx.compose.ui:ui-tooling-preview-android	1.11.2	uncertain/uncertain (0.180)	uncertain/uncertain (0.150)	0.150	0.150	none
commons-codec:commons-codec	1.15	uncertain/uncertain (0.151)	uncertain/uncertain (0.149)	0.149	0.000	1.16.0 (0.149), 1.16.1 (0.148)
androidx.datastore:datastore-preferences-android	1.2.1	uncertain/uncertain (0.160)	uncertain/uncertain (0.145)	0.145	0.145	none
androidx.datastore:datastore-core-android	1.2.1	uncertain/uncertain (0.152)	uncertain/uncertain (0.143)	0.143	0.143	none
com.fasterxml.jackson.core:jackson-databind	2.15.2	uncertain/uncertain (0.139)	uncertain/uncertain (0.142)	0.142	0.000	2.9.10.8 (0.142), 2.12.7.1 (0.141),

COMPONENT	SELECTED VERSION	IDENTITY	VERSION	SCORE	GAP	ALTERNATIVES
						2.13.5 (0.141)
androidx.viewpager2:viewpager2	1.0.0	uncertain/uncertain (0.132)	uncertain/uncertain (0.134)	0.134	0.134	none
androidx.startup:startup-runtime	1.1.1	uncertain/uncertain (0.114)	uncertain/uncertain (0.114)	0.114	0.114	none
androidx.compose.material:material-android	1.11.2	uncertain/uncertain (0.106)	uncertain/uncertain (0.109)	0.109	0.109	none

Vulnerabilities

No vulnerabilities were returned by the configured source, or vulnerability lookup was not run.

Native Libraries

PATH	ABI	SIZE	SHA-256
lib/arm64-v8a/libandroidx.graphics.path.so	arm64-v8a	10096	41e9a793c43a0f4fddb19e33f346bace464f30f888ba7b9eaf96294ea115bfb6
lib/armeabi-v7a/libandroidx.graphics.path.so	armeabi-v7a	7252	41399eba6fc2a60f6f14642375c1824f3cf25eb8fec7397d753730a3ceda3e2b
lib/x86/libandroidx.graphics.path.so	x86	9284	eb0570b41fd3bff25d8204a967c03bd7550719e768b791f680cc40cbe35f29af

PATH	ABI	SIZE	SHA-256
lib/x86_64/libandroidx.graphics.path.so	x86_64	10760	4e56c996f13670e70082658de7880c4020eabf4f25e43387f88ed78a713fc9f0

Artifacts

- Match details: `data/results/demo-osv-clean-fdroid/matches.json`
- CycloneDX SBOM: `data/results/demo-osv-clean-fdroid/sbom.cdx.json`
- JSON report: `data/results/demo-osv-clean-fdroid/report.json`
- Markdown report: `data/results/demo-osv-clean-fdroid/report.md`
- HTML report: `data/results/demo-osv-clean-fdroid/report.html`
- OSV results: `data/results/demo-osv-clean-fdroid/osv-vulnerabilities.json`

Limitations

- This MVP only detects libraries present in the local Maven reference DB.
- Heavy R8 optimization, reflection, packed APKs, encrypted DEX, and dynamic loading remain limited or unsupported.
- Native `.so` files are hashed, but native component version identification is not implemented in the first MVP.